

Counterexample to the arbitrary- D extension question

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2026-06-23

Source and Status

This packet concerns the final open question in Geivison Ribeiro's *Isometric embeddings of separable Banach spaces into $(\ell^\infty \setminus c) \cup \{0\}$* , arXiv:2508.18656 [1]. The question appears on page 9 of the arXiv PDF.

Open question. Let $D \subset \ell^\infty$ be an arbitrary (possibly nonseparable) subspace with $D \cap c = \{0\}$, and let E be a Banach separable. Does there always exist an isometry $T : E \rightarrow \ell^\infty$ such that $T(E) \cap (D + c) = \{0\}$?

Transcription of the Question

Let $D \subset \ell^\infty$ be an arbitrary, possibly nonseparable, subspace with $D \cap c = \{0\}$, and let E be a separable Banach space. Does there always exist an isometry $T : E \rightarrow \ell^\infty$ such that

$$T(E) \cap (D + c) = \{0\}?$$

Idea of the Proof

The statement allows D to be an arbitrary linear subspace, with no closedness, separability, or quotient-size restriction. A Hamel-basis complement to c is therefore admissible. For such a complement D , one has $D \cap c = \{0\}$, but also $D + c = \ell^\infty$. Thus every possible image of an isometry into ℓ^∞ is already contained in $D + c$, and a nonzero image cannot have trivial intersection with $D + c$.

Counterexample

Theorem 1. *The final open question in [1], as stated, has a negative answer.*

Proof. Let c denote the subspace of all convergent scalar sequences in ℓ^∞ . Since c is a proper linear subspace of ℓ^∞ , the standard Hamel-basis extension argument gives an algebraic complement D of c in ℓ^∞ . Equivalently,

$$\ell^\infty = D \oplus c.$$

In particular $D \cap c = \{0\}$, so D satisfies the hypothesis in the question. But $D + c = \ell^\infty$.

Now take the one-dimensional separable Banach space $E = \mathbb{K}$. If $T : E \rightarrow \ell^\infty$ is any isometry, then $T(E)$ is a nonzero one-dimensional subspace of ℓ^∞ . Since $D + c = \ell^\infty$, we have

$$T(E) \cap (D + c) = T(E) \neq \{0\}.$$

Thus no isometry $T : E \rightarrow \ell^\infty$ can satisfy the desired conclusion for this D and this separable E . $\square$

Remark 1. *The same D blocks every nonzero separable E , not only $E = \mathbb{K}$. The one-dimensional choice is enough to refute the universal quantifier in the source question.*

Limitations

This packet addresses the final arbitrary- D question only. It does not settle the earlier question in the source asking whether, for finite-dimensional D , one can always choose a closed nonseparable subspace $Z \subset \ell^\infty$ containing D and an isometric copy of a prescribed separable Banach space while still satisfying $Z \cap c = \{0\}$.

Verification and Novelty Check

The proof is purely algebraic and uses only the existence of a Hamel complement to a proper subspace. No computational verification is needed.

Before promotion, the run indexes were searched for arXiv:2508.18656, the source title, $D + c$, and related isometric-embedding phrases. Bounded web searches on 2026-06-23 used exact phrases from the open question and the algebraic-complement obstruction. No prior run packet or obvious published answer to this exact question was found.

Human review should mainly check that the intended reading of “subspace” in the final question did not silently require closedness or a proper quotient image. Under the literal wording, the algebraic complement is admissible.

References

- [1] G. Ribeiro, *Isometric embeddings of separable Banach spaces into $(\ell^\infty \setminus c) \cup \{0\}$* , arXiv:2508.18656, 2025.